

SECTION 12651**BUILDING MANGEMENT SYSTEM****Part 1 GENERAL****1.01 INTRODUCTION**

- A. Read and be governed by the Conditions of Subcontract, Schedule of Project Requirements and all other Sections of Division 1.

B. Definitions

In accordance with the Conditions of Subcontract and as specified herein.

Subcontractor means the MEP Installation Contractor.

BMS Controls Specialist – Specialist employed by Subcontractor for complete BMS installation works.

C. Coordination of the Works

In accordance with Section 12600: Submittals, Section 12700: Execution Requirements and as specified herein.

The Contractor and Subcontractor shall employ full time MEP coordinator(s) with their tasks dedicated to coordination issues.

The works described in this Section shall be fully coordinated in conjunction with all the Design Consultants drawings and specifications.

1.02 PRIORITY OF DOCUMENTS

- A. Refer to the Conditions of Subcontract

1.03 SCOPE OF WORK

- A. The following specification details Building Management System (BMS) and shall be applied in all cases when work specified under the relevant Sections of Technical Specifications require and enables connection to the BMS.

- B. The Building Management System (BMS) to be provided shall perform a series of general functions: This includes, but is not limited to, the following:
1. Building Management and Control
 2. Monitoring and Control of Controllers, Remote Devices and Programmable Logic Controllers
 3. Operator Interface
 4. Video display integration
 5. Data collection and Historization
 6. Alarm Management
 7. Trending
 8. Report Generation
 9. Network Integration
 10. Data exchange and integration with a diverse range of other computing and facilities systems using industry standard techniques.
- C. The Subcontractor shall be responsible for submitting the above works, complete furnishing and installing a fully integrated building automation system, incorporating direct digital control (DDC) for energy management, equipment monitoring and control, and subsystems as herein specified. The system shall include all computer software and hardware, operator input/output devices, sensors and controls required for complete operation. The above scope shall be based on Design Consultants approvals of submitted samples and existing standards of works.
- D. The Subcontractor shall be responsible for all BMS wiring for a complete and operable system. All wiring shall be performed in accordance with all local and international codes and standards of works.
- E. The Building Management System (BMS) herein specified shall be fully integrated and installed as a complete package by the Subcontractor. The System shall include all wiring, piping, installation supervision, calibration, adjustments, and checkout necessary for a complete and fully operational system.
- F. Installation of Building Management System (BMS)
The Subcontractor shall furnish and install a complete Building Management System (BMS) for all mechanical systems and other facility systems as included in the project documents. The BMS will

provide the functional features as defined in Part 1-General, Part 2-Products, and Part 3-Execution of these Specifications.

G. The Subcontractor shall provide a complete and operational system to perform all sequences of operations stated within Section 3 or as shown on the control drawings.

H. In addition, the following apply:

1. The work under this Section shall include all materials and labour to perform all work required for the installation of the BMS as specified.
2. The drawings and Specifications are complementary to one another—meaning that what is called for on one is to be considered called for in both. Where conflicts exist, the more stringent requirement shall apply.
3. The Subcontractor shall be responsible for field verification of site conditions and for gathering all necessary field data for all items to be provided under this contract prior to submitting his or her bid.
4. Where work specified under other Sections of this Specification connects to equipment or systems that are listed and described in this Section, the Subcontractor shall provide proper connection(s) to such equipment, including trade coordination.

I. Related Work Specified Elsewhere:

Section 12600	Submittals
Section 12700	Execution Requirements

DIVISION 14 Special Constructions

(Particularly relevant are following sections from the division 13):

Section 14851, FIRE DETECTION AND ALARM SYSTEM.

Section 14860, CARBON MONOXIDE DETECTION AND ALARM

Section 14920, FIRE PUMPS

DIVISION 15 Conveying Systems

DIVISION 16 Mechanical

(Particularly relevant are following sections from the division 15):

Section 16010, BASIC MECHANICAL REQUIREMENTS.

Section 16070, MECHANICAL SOUND, VIBRATION & SEISMIC CONTROL

Section 16110, VALVES

Section 16130, PUMPS

Section 16440, PLUMBING PUMPS

Section 16450, POTABLE WATER STORAGE TANKS

Section 16460, DOMESTIC WATER TREATMENT

Section 16720, AIR HANDLING UNITS

Section 16760, TERMINAL HEATING AND COOLING UNITS

Section 16820, DUCT ACCESSORIES

Section 16830, FANS

Section 16900, HVAC INSTRUMENTATION AND CONTROLS

Section 16940, SEQUENCE OF OPERATION

Section 16950, TESTING ADJUSTING AND BALANCING

DIVISION 17 Electrical

1.04 SUBMISSIONS

- A. Provide submittals in accordance with the Conditions of Subcontract, Project Record Documents, Administrative Procedures, Product Requirements and as specified herein
- B. Samples of all types of room mounted equipment (i.e. detectors, thermostats, etc.) shall be provided by the BMS/Controls Specialist for approval by the Design Consultant.
- C. Each submittal shall include the following information.

1. A section by section listing of clarifications and exceptions taken to these specifications.
 2. BMS riser diagram showing all DDC controllers, operator workstations, network repeaters, and network wiring.
 3. One-line schematics and system flow diagrams showing the location of all control devices.
 4. Points list for each DDC controller, including: Tag, Point Type, System Name, Object Name, Expanded ID, Display Units, Controller Type, Address, Cable Destination, Module Type, Terminal ID, Panel, Slot Number, Reference Drawing, and Cable Number.
 5. Product information sheets with identification by the manufacturer's part number for each model of BMS controller identified on the BMS Riser Diagram to be provided.
 6. Catalogued cut sheets of all equipment used. This includes, but is not limited to, the following: DDC panels, peripherals, sensors, actuators, dampers, control air system components, and so forth.
 7. Range and scale information for all transmitters and sensors. This sheet shall clearly indicate one device and any applicable options. Where more than one device to be used is on a single sheet, submit two sheets, individually marked.
 8. The sequence of operations shall cover normal operation and operation under the various alarm conditions applicable to that system.
 9. Points shall be referenced by name, including all software points such as programmable set points, range limits, time delays, and so forth.
- D. Project organization including resumes of each project team member.

- E. User interface functional outline. The outline shall include each display screen to be provided, data to be displayed, and links to other screens. The outline level hierarchy shall be:
 - 1. Site
 - 2. Building
 - 3. Floor
 - 4. System
- F. Detailed Bill of Material list for each panel, identifying: quantity, part number, description, and associated options.
- G. Control Damper Schedules. This spreadsheet type schedule shall include a separate line for each damper and a column for each of the damper attributes, including: Code Number, Fail Position, Damper Type, Damper Operator, Blade Type, Bearing Type, Seals, Duct Size, Damper Size, Mounting, and Actuator Type.
- H. Control Valve Schedules. This spreadsheet type schedule shall include a separate line for each valve and a column for each of the valve attributes, including: Code Number, Configuration, Fail Position, Pipe Size, Valve Size, Body Configuration, Close off Pressure, Capacity, Valve CV, Calc CV, Design Pressure, Actual Pressure, and Actuator Type.

1.05 QUALITY CONTROL AND ASSURANCE

- A. In accordance with the Conditions of Subcontract, Section 01400: Quality Control and as specified herein.
- B. Submit the prior project list of five Building Management Systems of similar size, scope and complexity, which have been completed and accepted by the Sub-consultant. Include, Project Name, Owner Name and Address, Contact Name, Contact Phone Number and a brief description of the system. These systems must be online and functional.
- C. Location of the BMS Specialists branch office that would perform both routine maintenance and emergency repair services. Include resumes of each hardware and software technician who would be involved in performing on-site service.

- D. The BMS Specialist shall be a factory owned branch office that is regularly engaged in the engineering, programming, installation and service of Building Management Systems of similar size and complexity. Emergency service shall be available on a 24-hour, 7-day-a-week basis.
- E. The Subcontractor shall be responsible for all coordination of work in a satisfactory and neat workmanlike manner acceptable to the Design Consultant.
- F. The Subcontractor will coordinate with other Trade Subcontractors regarding the location and size of pipes, equipment, fixtures, conduit, ducts, openings, switches, outlets.
- G. Experience Record: The BMS Specialist shall have a minimum of ten years experience with the complete, installation of Building Management Systems of similar size and technical complexity.
- H. The Subcontractor shall employ specialists in the field of Building Management Systems including: Programming, Engineering, Field Supervision, and Installation. Specialists shall have a minimum of five years of experience with Building Management Systems.
- I. Controllers and DDC (Direct Digital Control) system components shall be current production products.
- J. All other equipment shall be the products of the BMS manufacturers or of an approved manufacturer regularly engaged in production of specialized Building Management System materials or equipment.
- K. ISO-9001: The manufacturer and software developer of the Building Management System shall provide documentation supporting compliance with ISO-9001 (Model of Quality Assurance in Design/Development, Production, Installation, and Servicing). Product Literature provided by the BMS manufacturer shall contain the ISO-9001 Certification Mark from the applicable registrar. Manufacturers delivering products that do not comply with the ISO-9001 certification requirement shall provide the following information to assure that quality systems are in place, which are equivalent to the ISO-9001 standard:

1. Marketing Specification Standards
2. Design File Standards
3. Manufacturing Test Standards
4. Calibration Standards
5. Quality System Standards
6. Quality System Procedures
7. Documented management commitment that all employees participate in quality programs
8. Training Procedures
9. Methods by which corrective actions are taken for problems identified within the factory process.

L. Governing Code Compliance

The Subcontractor shall comply with all current governing codes ordinances and regulations, including local regulations, UL, NFPA, , NEC, and so forth.

M. FCC Regulation

All electronic equipment shall conform to the requirements of FCC Regulation, Part 15, Section 15, Governing Radio Frequency Electromagnetic Interference, and be so labelled.

1.06 DELIVERY, STORAGE AND HANDLING

- A. In accordance with the Conditions of Subcontract, Section 01600: Materials and Equipments.

1.07 SPARE PARTS

- A. In accordance with the conditions of Subcontract, Section 01800: O&M Manuals and as specified herein.

1.08 WARRANTIES

- A. In accordance with the Conditions of Subcontract, Section 12600: Submittals, Section 12700: Execution Requirements, Section 12800: O&M Manuals.

B. Material:

The Control System component, system software, parts and assemblies shall be free from defects in material and workmanship under normal use and service. If within thirty six (36) months from the date of completion any of the equipment herein described is defective in operation, workmanship or materials, it will be replaced, repaired or adjusted at the option of the Subcontractor.

C. All corrective software modifications made during warranty service periods shall be updated on all user documentation and on user and manufacturer archived software disks.

1.09 MAINTENANCE

A. In accordance with the Conditions of Subcontract, Section 12600: Submittals, Section 12800: O&M Manuals.

1.010 INSPECTIONS AND TESTS ON COMPLETION AND COMMISSIONING

A. In accordance with the Conditions of Subcontract, Section 01400: Quality Control, Section 01457: Inspection and Testing Control, Section 12800: O&M Manuals.

B. A copy of the tests and commissioning report of the equipment/system shall be bound into the O&M Manual.

1.011 OPERATION AND MAINTENANCE MANUAL AND TRAINING OF PERSONNEL

A. In accordance with the Conditions of Subcontract, Section 01800: O&M Manuals and as specified herein.

B. O&M manuals shall be prepared in accordance to BSRIA application guide 1/87.1 "Operating and maintenance manuals for building services installations" and shall include, but not be limited to, the following categories (See also additional requirements of Section 12800):

1. Contractual and legal guides
2. The purpose of the installation
3. Installation records
4. Description of the installation;
5. Equipment schedule
6. Parts identification and recommended spares
7. Commissioning data;

8. Operation and how installation is to be used and keep operational;
 9. Maintenance schedules;
 10. How installation may be modified or changed;
 11. Disposal of installation;
 12. Names and addresses of manufacturers;
 13. Set of "as built drawings"
 14. Emergency Operating Procedures
 15. Manufacturers literature;
- C. Provide copies of all manufacturers manuals covering the installed system. This shall include, as a minimum:
1. System Engineering Manual and Hardware Manual
 2. System Installation Manual
 3. Programming Manual
 4. Engineering and Troubleshooting Bulletins
 5. Operator Workstation Software Manual
 6. All other pertinent manuals published by the control system manufacturer.
- D. Include the following documentation in the Hardware Manual:
1. General description and cut sheets for all components.
 2. Detailed wiring and installation illustrations and complete calibration procedures for each field and panel device.
 3. Complete trouble-shooting procedures and guidelines.
 4. Complete operating instructions for all systems.
 5. Maintenance Instructions: Document all maintenance and repair/replacement procedures.
- E. Include the following documentation in the DDC Software Manual:
- Sequence of Operations
 - Program Listing of Software Source Code OR Flow Chart Diagrams of Programming Objects.
 - Printed listing of controller and operator workstation database files.
 - Software Point Name Abbreviation List.. Include Name, Description, Controller Where Located, Point Type and Point ID.
 - I/O Point List. Include Point Name, Controller Location, Point Number, Control Device, Range and Span.
 - Printouts of all; Reports, Group Listings and Alarm Messages.
 - Index of all DDC point names with documentation manual page number references.
- F. Software manuals for all applications programs to be provided as a part of the operator workstations, portable operator terminals, programming

devices, and so forth for evaluation for compliance with the performance requirements of this Specification.

G. Submit all Record Drawings.

H. The services of a factory trained, field service representative shall be provided to supervise the final testing, charging, initial start-up and conduct concurrent operator instruction where applicable.

I. Training

Upon completion and acceptance of Installation, properly acquaint Representative of the Employer with details and sufficient training to enable them to obtain maximum efficiency in the system operation.

1.012 DESIGN CRITERIA, REGULATIONS AND STANDARDS

A. Refer to Section 01400: Quality Control, Section 15010: Basic Mechanical Requirements and as specified herein.

B. All works shall be compliant with the requirements of Municipality, local authorities and utility companies.

Part 2 PRODUCTS**2.01 DEFINITIONS (TECHNICAL)**

- A. Algorithm: A software procedure for solving a recurrent mathematical or logical problem.
- B. Analogue: A continuously varying signal or value (temperature, current, velocity, etc.).
- C. Binary: A two-state system where an "ON" condition is represented by a high signal level and an "OFF" condition is represented by a low signal level.
- D. BMS Building Management System, BMCS Building Management and Control System and Facility Management System (FMS):
 - 1. BMS as used within this Section of Specifications is synonym for both BMCS and FMS being preferred by some. BMS also is the entire system of hardware and software specifically designed to centrally manage building HVAC and related utilities as well as other specified systems and utilities. The BMS (BMCS, FMS) includes the DDC subsystem, open system ports, and open protocol bus or integrators and network routers for connection to information networks.
- E. Control Process: The software required to perform a complete control loop from input signal to interlock logic, process calculation to final output signal control.
- F. Control Wiring: Includes conduit, wire and wiring devices to install a complete Control System including motor control circuits, interlocks, thermostats, PE and EP switches and like devices. Includes all wiring from a DDC cabinet to all sensors and points defined in the Points List summary or specified herein and required to execute the sequence of operation. Includes necessary power wiring to all BMS devices, digital controllers including terminal units and actuators.
- G. Deadband: A temperature range over which no heating or cooling energy is supplied, such as 22-24 degrees C, i.e. as opposed to single point changeover or overlap, or a range from setpoint over which no control action is taken.

- H. Direct Digital Control System: The portion of the BMS which provides closed loop control of all HVAC equipment.
- I. Distributed Control: A system whereby all control processing is decentralized and independent of a central computer. The control system is built up of stand-alone controllers. A single controller failure shall not impact more than one system.
- J. Integration: The ability of control system components from different manufacturers to connect together and provide coordinated control via real-time data exchange through a common communications data exchange protocol. Integration shall extend to the operator's workstation software, which shall support user interaction with all control system components. Methods of integration include industry standard protocols such as: BACnet, LonMark/LonTalk, OLE for Process Control (OPC) or integrator interfaces between cooperating manufacturer's systems.
- K. Network: A system of distributed control units that are linked together on a communication highway. A network allows sharing of point information between all control units. Additionally, a network provides central monitoring and control of the entire system from any distributed control unit location. First tier networks shall provide "Peer-to-Peer" communications. Second tier networks shall provide either "Peer-to-Peer", Master-Slave or Supervised Token Passing communications.
- L. Open Protocol Bus (OPB): A pre-programmed communications integrator that allows devices from one manufacturer to communicate and interact with those of another.
- M. Open System Port (OSP): A user programmable communications port that provides the ability to develop custom communications processes to integrate other operating systems with the BMS System.
- N. Operator-Machine Interface: A method by which an operator communicates with a BMS System. Operator-machine interfacing allows an operator to command, monitor, and program the system.
- O. Peripheral: Input/Output equipment used to communicate with the computer and make copies of system outputs, peripherals include CRT, printer, tape deck, diskette, etc.

- P. Pick Point: A pick point is a graphical display element that allows the operator to “click” the item and automatically display the associated screen or service. Any screen may have pick points to or be linked from any other screen. Pick points shall be configured on each display screen to provide a logical user navigation system using a ladder tree hierarchy.
- Q. PID Control Loop: A mathematical calculation used to evaluate a control input and determine the control output value required to maintain the input value at setpoint. The PID (Proportional, Integral, Derivative) control loop shall have operator adjustable maximum rate of change, P and D gains and loop response time delay. The loop shall be self-integrating so that no integral constant is required and the loop shall not be subject to “Integral Windup”.
- R. The term Provide within this Section shall mean “Provide complete in place”, that is, furnished and installed and ready for operation and use.

2.02 BMS HARDWARE – GENERAL

- A. The control system shall be furnished and installed and shall consist of a PC-based Operator Work Station (OWS) with colour-graphic data display as specified within this Section, microcomputer-based DDC panels interfacing directly with sensors, actuators and environmental delivery systems. The system shall be a true distributed processing system. All software control functions shall be performed by intelligent field panels, or Network Control Units (NCUs). The control system shall be digital electric/electronic with electrically actuated valves and dampers.
- B. The BMS shall possess a fully modular architecture, permitting expansions through the addition of more standalone control units, sensors, actuators, or operator terminals. Expansion beyond this capacity must be able to be done in modules without abandoning any initial equipment.
- C. The BMS system shall provide maximum flexibility responding to changes in the buildings use, supporting the use of BACnet, LON, EIB, Profibus and Ethernet TCP/IP, Token Ring/Telephone network and Internet communication technologies.
- D. All materials and equipment used shall be standard components kept in inventory by the BMS manufacturer, regularly manufactured for this and/or other systems and not custom designed specifically for this

project. The components shall not require customisation other than setting jumpers and switches, adding firmware modules or software modules or software programming to perform required functions. All systems and components shall have been thoroughly tested and proven in actual use for at least two (2) years. Products having less than a two-year field service record will be acceptable if a certified record of satisfactory field operation, for not less than 6,000 hours exclusive of the manufacturer's factory tests, can be shown.

- E. The BMS, except for controls specified to be furnished as components of the mechanical equipment, shall be provided as herein specified and as required to produce complete and properly functioning controls for all systems described in the Section 15900, HVAC Instrumentation and Controls and Section 15940, Sequence of Operation.
- F. The Building Management System (BMS) system shall use a Client Server architecture based around a modular PC network, utilising industry standard operating systems, networks and protocols.
- G. The system shall allow the distribution of system functions such as monitoring and control and graphical user interface etc. across the network to allow maximum flexibility and performance. The architecture shall include support of various Wide Area Networks using standard hardware and software to link nodes into a single integrated system. The network protocol used shall be industry standard TCP/IP. The system shall also support remote configuration and operation using standard dial-up modems.
- H. The BMS system shall allow communications with a wide variety of control devices utilising off the shelf driver packages.

2.03 SERVER HOT BACKUP

- A. This facility shall enable the system server to operate in high availability architecture with no single point of failure. To achieve this, the facility must provide specific features.
- B. The system must be capable of running a pair of similarly configured computers in a hot backup configuration where at any point in time, one is the acting Primary and the other the acting as the Hot Backup. An on-line database duplication mechanism must be supported.

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- C. Simply scanning I/O on two separate systems and processing independently is not acceptable. The database duplication must be performed on a per-transaction basis for two reasons:
 - 1. To ensure that the duplicated Backup database is consistent at all times with the Primary database
 - 2. To avoid unnecessary loading of field devices caused by duplicate polling
- D. It must be possible to remove one of the redundant systems for maintenance without interrupting operation, and upon its reinstatement, re-synchronise the databases, again without interruption to system operation. A method of manually initiating a fail-over must be provided to assist with such maintenance operations.
- E. Failure of either system must be announced audibly and visually via the alarming subsystem.
- F. To accommodate recoverable faults, the failed system must be able to reboot automatically after non-fatal errors and assume the role of acting as Hot Backup automatically.

2.04 COMMUNICATIONS REDUNDANCY

- A. The system must be capable of supporting fully duplicated communications links to Operator Workstations and field devices that support this type of connection.
- B. Capabilities must exist to interface to devices via:
 - 1. Direct Serial Connection (or modem)
 - 2. Serial Connection via Ethernet connected terminal servers. Using back-to-back terminal servers will be deemed unacceptable. The system must be able to directly communicate with the Terminal servers via Ethernet.
- C. The system and its associated Operator Workstations must be capable of connecting to two fully independent Ethernets run in parallel. No repeater or bridge connection between the Ethernets is acceptable as a means of achieving this function.
- D. Operator Workstations must be capable of switching automatically between the two server computers in the event of a fail over, and switching between two Ethernets automatically in the event of an Ethernet failure.

- E. The architecture of the Building Management System shall consist of three levels:
1. The Field Level
 2. The Automation Level
 3. The Management Level

2.05 THE FIELD LEVEL

A. General

The Field Level shall consist of a distributed network of DDC Controllers, FCU Controllers, Protocol Translators and Factory Mounted Controllers.

The functions of the DDC Controllers, FCU Controllers and Factory Mounted Controllers are:

- Physical connections to field devices such as sensors, actuators, transducers and relays
- Conversion of analogue inputs from a physical variable to an engineering unit such as converting an input of 0-10VDC or 4-20mA to engineering units of 0-100%RH or 0°C - 50°C
- Conversion of analogue outputs from engineering units to physical variables such as 0-100% to a 0-10VDC signal
- Reporting changes from field devices to the automation level
- Translating commands from the automation level to output signals
- Local control functions such as PID control loops, interlocks, local calculations and execution of default start/stop time schedules

- B. DDC Controllers and FCU Controllers shall be provided under this contract. Factory Mounted Controllers shall be provided by others. The Protocol Translators required to integrate the Factory Mounted Controllers shall be provided under this contract. Protocol Translators shall be provided to integrate with various other subsystems, equipment mounted controls etc. as described elsewhere in the tender documents

- C. Each DDC Controller and FCU Controller shall operate as a stand-alone controller capable of performing its specified control responsibilities, including PI loop control, independently of other controllers in the network.

- D. For fault tolerance, a dedicated FCU Controller shall be provided for each FCU. It is the intent that the failure of any one Controller not affect the operation of more than one FCU

- E. For fault tolerance, it is not permitted to connect, directly or indirectly, more than 48 physical points to any DDC Controller. It is the intent that the failure of any one DDC Controller not affect more than physical 48 points. Physical points are defined as analogue inputs, analogue outputs, binary inputs and binary outputs. Data points stored in the DDC Controller are not considered physical points.

2.06 GENERAL PURPOSE CONTROLLERS AND STANDALONE DIGITAL CONTROL UNITS

- A. General Purpose DDC Controllers and Standalone Digital Control Units (SDCU) shall provide stand alone control of HVAC and lighting. Each controller shall have its own control programs and shall continue to operate in the event of a failure to its associated Ethernet or Remote Site controller.
- B. Memory: Control programs shall be stored in battery backed-up RAM and EPROM. Each controller shall have a minimum of 32K bytes of user RAM memory and 128K bytes of EPROM.
- C. Communication Ports: SDCU's shall provide a communication port to the field bus. In addition, a port shall be provided for connection to a laptop computer to support local programming and parameter changes. It shall be possible from this port to access and program any controller on the field bus, any Ethernet or Remote Site controller, or any SDCU on a different field bus.
- D. Input/Output: Each SDCU shall support the addition of the following types of inputs and outputs:
 - 1. Digital Inputs for status/alarm contacts
 - 2. Counter Inputs for summing pulses from meters.
 - 3. Thermistor Inputs for measuring temperatures in space, ducts and thermowells.
 - 4. Analogue inputs for pressure, humidity, flow and position measurements.
 - 5. Digital Outputs for on/off equipment control.
 - 6. Analogue Outputs for valve and damper position control, and capacity control of primary equipment.

- E. Expandability: Input and output capacity shall be expandable through the use of plug-in modules, minimum of two modules shall be added to the base SDCU before additional power is required.
- F. Networking: Each SDCU shall be able to exchange information on a peer to peer basis with other standalone digital control units during each field bus scan. Each SDCU shall be capable of storing and referencing global variables. Each SDCU shall be able to have its program edited and/or modified either locally through a laptop computer or through a workstation connected to an Ethernet control unit.
- G. Indicator Lamps: SDCU's shall have as a minimum, LED indication of CPU status, and field bus status.
- H. Real Time Clock (RTC): An SDCU shall have contain a real time clock in either hardware or software. The accuracy shall be within 10 seconds per day. The RTC shall provide the following information: time of day, day, month, year, and day of week. Each SDCU shall be capable of receiving a signal over the network from the Controller to synchronize all SDCU real time clocks to the same time.
- I. Power Supply: The SDCU shall operate at 24 VAC (50/60 Hz) with an accuracy of +/- 20%.
- J. Automatic Restart After Power Failure: Upon restoration of power, the SDCU shall automatically and without human intervention: update all monitored functions; resume operation based on current, synchronized time and status, implement special start-up strategies as required.
- K. Battery Back Up: Each SDCU shall have at least 3 years of battery back-up to maintain all volatile memory.
- L. Alarm Management: For each system point, alarms can be created based on high/low limits or conditional expressions. All alarms shall be tested each scan of the SDCU and can result in the display of one or more alarm messages or reports.

1. Up to 8 alarms can be configured for each point in the controller.
 2. Alarm messages can be sent to a local terminal or modem connected to a Controller or RSCU, or to the Operator Workstation.
 3. Alarms shall be generated based on their priority. A minimum of 255 priority levels shall be provided.
 4. If communication with the Controller or RSCU is temporarily interrupted, the alarm shall be buffered in the SDCU. When communications return, the alarm shall be transmitted to the Controller or RSCU if the point is still in the alarm condition.
- M. The DDC Controller shall be UL or CE listed. Each DDC Controller shall be installed with a minimum of 10% spare point capacity such that additional points may be connected by adding an end device, wiring and programming.
- N. The DDC Controller shall be designed to be mounted in a mechanical room environment with a rated operating temperature range of 0°C to 45°C. If the proposed DDC Controller is not rated to 45°C, DDC enclosure shall be equipped with a fan for ventilation.
- O. Each DDC Controller shall be equipped with a battery-backed real-time clock. The intent is that if communications with the system controller is interrupted then the DDC Controller is still capable of using a default seven-day start/stop schedule.
- P. The required control programs for each DDC shall be stored in a non-volatile (EEPROM) memory
- Q. The DDC Controller shall be provided with active RAM independent of EEPROM, which shall be battery-backed.
- R. Each DDC Controller shall have a dedicated service port to plug in the portable operator's terminal (POT). It shall be possible to read, write and change any parameters on a bus by plugging the POT to any one of the Controllers on the communication trunk. The portable operator's terminal shall have a visual and audible alarm with mute facility on its fascia. The POT shall have minimum three password levels. Separate cabling for connecting the POT shall not be acceptable. Changing the parameters locally from any Outstation shall be done by POT which is truly portable and hand held and not via Laptops.

- S. The DDC Controllers shall be fully programmable to meet the sequence of operation and future modifications. For ease of support and documentation, the preferred method of programming is graphical.
- T. Each DDC Controller shall be programmable with a graphical language which includes as a minimum, the following modules:
1. Setpoint jump
 2. Positioning time
 3. P-controller (reverse or direct acting)
 4. P1-controller (reverse or direct acting)
 5. PI-controller with I-deletion (reverse or direct acting)
 6. PID controller (reverse or direct acting)
 7. 2-point controller (reverse or direct acting)
 8. Proportional additional sequences (reverse or direct acting)
 9. Data transmitter (digital or analogue)
 10. Data converter (analogue-digital or digital-analogue)
 11. Ring Counter
 12. Timer (switch on or switch off)
 13. Logic operations:
 - logic "AND" (2,3 or 4)
 - logic "OR" (2,3 or 4)
 - logic "EXOR"
 - logic "NOT"
 14. Comparative operations:
 - Maximum values (2,3 or 4)
 - Minimum values (2,3 or 4)
 - Average values (2,3 or 4)
 15. Enthalpy calculation
 16. Optimiser
 17. Mean value calculation
 18. Hysteresis
 19. Output steps (digital or analogue)
 20. Digital outputs shall be potential free outputs. Analogue outputs shall be true analogue outputs (0-10 V DC, 0-20 V DC, 0-20 mA & 4-20 mA)
- U. Above blocks shall be resident in the DDC Controllers and independent of any high level interfaces/controllers.

- V. Further, the DDC unit software must have the following additional functions:
1. Free selection of range and unit (dimension) of all signals (measured values, accumulated values, calculated values, etc.)
 2. Free allocation of access protection in accordance with operating priorities
 3. Free definition of manual override priorities (software) from operator terminal and/or management station.
- W. The Protocol Translator software shall allow information from the Factory Mounted Controllers to be accessed by the automation level. Physical points and data points from the Factory Mounted Controllers shall be mapped into the data base in the System Controllers. The Protocol Translator shall manage all communication handshaking necessary with the Factory Mounted Controllers. The Protocol Translator shall support Analogue Inputs, Analogue Outputs, Analogue Data, Binary Inputs, Binary Outputs and Binary Data as provided by the Factory Mounted Controllers.
- X. Air Handler Controllers (To be used on units with less than 40 points)
1. AHU Controllers shall be capable of meeting the requirements of the sequence of operation found in the Execution portion of this specification and for future expansion.
 2. AHU Controllers shall support all the necessary point inputs and outputs as required by the sequence and operate in a standalone fashion.
 3. AHU Controllers shall be fully user programmable to allow for modification of the application software.
 4. A LCD display shall be optionally available for readout of point values and to allow operators to change set points and system parameters.
 5. A manual override switch shall be provided for all digital and analogue outputs on the AHU Controller. The position of the switch shall be monitored in software and available for operator displays and alarm notification.

2.07 THERMAL ENERGY MEASUREMENT

- A. Tenants shall be billed for the air-conditioning utility based on their thermal energy consumption and not on "carpet area" basis. This shall be achieved as follows:

- B. Each of the secondary Chilled water circuits (supply and return pipelines) shall have temperature sensors and Thermal Energy Flow meter to ascertain volumetric flow.
- C. Based on the temperature difference between flow and return chilled water and the volumetric flow, the thermal energy delivered to the tertiary circuits by the chillers shall be computed.
- D. Electrical power meters for each chiller, primary Ch. W. and Condenser pump together with total of the consumption of the Cooling tower fans shall ascertain power consumed by these machines and therefore electrical costs towards generating air-conditioning energy.
- E. Thermal energy consumed by each tertiary system shall thus be computed and
- F. Individual Tenants including each of Apartments shall have sensors mounted on the main supply and return chilled water line for each tenant. Thermal Energy Meter shall be located in the return line. The meter shall calculate the thermal energy consumed by all terminal units (set of FCUs and AHU-s).
- G. This arrangement facilitates the correct billing to each tenant based on the thermal energy consumed by the tenant. (Software shall allow application of a multiplying factor to take care of common utilities such as Fresh Air AHU / Corridor/Lift lobby air-conditioning , Extract ventilation etc.,.)
- H. Electrical meters to the incoming DB of each tenant shall measure electrical power consumed. Domestic water meters shall be used, one per tenant, to measure the amount of domestic water consumed.
- I. The building Maintenance Company shall produce a bill, which summates the total bill a tenant has to pay, viz., Thermal energy costs.
- J. Individual meters shall have a data communication port. These shall be linked via a data communication cable. Information available from these meters shall be transferred to a central unit via this communication bus (the M-bus). The M-bus shall be connected to the Central Building management System via suitable interfaces

K. Description of Thermal Energy Meters:

Flow meters shall work in conjunction with a pair of temperature sensors. Flow meters shall have in-built algorithm to compute the thermal energy consumed based on the volumetric flow and the temperature difference. Flow meters shall have display unit to indicate as a minimum:

- Supply temperature
- Return temperature
- Temperature difference
- Instantaneous Flow
- Instantaneous thermal Energy

1. The display units shall be remotely connected to the flow-measuring device. The display unit shall be placed at an easily accessible location. The flow - meter shall have in-built fault diagnostic features for easily trouble-shooting.
2. Appropriate modules for data communication from the flow meter for remote monitoring / billing shall be part of the offer.

2.08 APPLICATION SPECIFIC CONTROLLERS (ASC)

A. Application specific controllers shall be used for terminal devices such as Fan Coil Units and the like. The ASCs shall be LONMARK products communicating on LON bus. Any failure problem in communication bus should not affect the working of the ASC controller. A dedicated stand-alone controller shall be provided for each FCU box. A common controller for FCUs serving different areas shall not be acceptable. These controllers shall be looped with a bus cable and connected to the automation level via an interface unit. The apartment FCU shall be connected to the BMS or shall be stand alone as indicated on drawings.

B. In general ASC shall comply with the following specifications

- For FCUs
- PID control
- Downloadable application software
- LonMark compatible bus communication

- To be integrated to the management station software
 - Control of AC 24 V PWM valve actuators, 3 point AC 24 V valve and damper actuators, or electric heating coils
 - Volt-free relays for fan control
 - Operating Voltage → 220 V
 - Internal fuse, thermal, automatic reset
 - Connectable to 1 room unit via PPS2, 4 wire unscreened twisted pair @ 4.8 KBPS
- C. The application specific controllers shall be capable of working in conjunction with the following type of room controllers. The specific type of room controller to be used in specific applications shall be selected from any one of the following types to meet the description written in the sequence of operation.
1. TYPE 1
 - Integrated room temperature sensor
 2. TYPE 2
 - Integrated room temperature sensor
 - Dial for temperature setpoint
 3. TYPE 3
 - Integrated room temperature sensor
 - Dial for temperature setpoint
 - Rocker switch for off/auto mode (single speed fan)
 4. TYPE 4
 - Integrated room temperature sensor
 - Dial for temperature setpoint
 - Rocker switch for off/auto mode and fan speeds (3 speed fan)
 5. TYPE 5
 - Integrated room temperature sensor
 - Dial for temperature setpoint
 - Rocker switch for off/auto1 mode and fan speeds (3 speed fan)
 - LCD display of measured temperature
 6. TYPE 6
 - Integrated room temperature sensor
 - Rocker switch for temperature setpoint (raise/lower)
 - Rocker switch for off/auto1 mode and fan speeds
 - LCD display of measured temperature
 - Communication with controllers via Lon bus
 - Exchangeable rocker switches for lighting and blinds
 - Selection of downloadable software applications for the operation of lighting and blinds

- Operating mode 'auto' –comfort, 'off' – standby or economy
- D. It shall be possible for the FCU controller to automatically change the FCU motor speed based on the temperature deviation. The FCU shall continue to operate at the low speed when the temperature conditions are achieved. It shall be possible to switch OFF the FCU motor either from the ON/OFF switch of the room unit or via BMS PC.
- E. The FCU controller shall modulate the modulating valve to meet the desired temperature condition.
- F. In case multiple FCUs are serving the same common area a common controller can be used if suitable for the site conditions.
- G. Unitary Controllers
1. Unitary Controllers shall support, but not be limited to, the control of the following systems as described in the Execution portion of this specification, and for future expansion:
 - a. Fan Coil Units
 - b. Ventilation Units etc.
- H. Unitary Controller I/O
1. The I/O of each Unitary Controller shall contain the sufficient quantity and types as required to meet the sequence of operation found in the execution portion of this specification. In addition, each controller shall have the capability for time of day scheduling, occupancy mode control, after hours operation, lighting control, alarming, and trending.
- I. Lighting Controllers
1. Lighting controllers shall provide direct control of 20 Amp, 240 VAC lighting circuits using mechanically held, latching relays. Controllers shall contain from 8 to 48 circuits per enclosure. Each controller shall also contain inputs for direct connection to light switches and motion detectors.

2. Each controller shall have the capability for time of day scheduling, occupancy mode control, after hours operation, alarming, and trending.

J. Display Controllers

1. Display controllers are standalone, touch screen based operator interfaces. The controller shall be designed for flush mounting in a finished space, with a minimum size of 9 x 9 inches.
2. Software shall be user programmable allowing for custom graphical images that simulate floor plans, menus, equipment schematics along with associated real time point values coming from any SDCU or Controller on the network.
3. The touch screen display shall contain a minimum of 64 possible touch cells that permit user interaction for changing screens, modifying set points or operating equipment.
4. Systems that do not offer a display controller as specified must provide a panel mounted computer with touch screen capability as an alternative.

2.09 THE FIELD LEVEL COMMUNICATIONS NETWORK

- A. The DDC Controllers, ASC Controllers and Protocol Translators shall be connected on a communications network, which shall be supplied under this contract.
- B. The DDC controllers must ensure the communication upward to the automation level and on to the same level to the other DDC Controllers (peer to peer communication). This peer to peer communication between DDC Controllers on the bus shall be independent of the higher level system controllers
- C. The communications network for DDC Controllers shall use standard EIA-485 signalling and for ASC Controllers network communication shall be LONworks.

2.010 THE AUTOMATION LEVEL

- A. The automation level shall consist of a distributed network of System Controllers.
- B. The system controller of the BMS shall be an intelligent module that can be operated autonomously and does not need to be located centrally. In information technology terms, it may be described as a network node.

- C. The System Controller shall co-ordinate the flow of data in both directions via three buses: it shall be responsible for the communication upwards to the management station, on the same level with other system controllers and downwards via the interface to the locally operating DDC/ASC Controllers.
- D. The System Controller shall contain a comprehensive range of function blocks, with which digital and numerical variables can be logically connected in a wide variety of configurations. Programming shall use block technology (on the same principles as those used in functional block diagram technology).
- E. The System Controller shall be capable of sorting a wide variety of data (e.g. alarm and error messages) into categories based on any criteria. The data shall be prepared in such a way that it can be used directly by the management station, a local printer or a local operator terminal.
- F. To implement the energy saving programmes function blocks like optimum start-stop, duty cycling, etc., should be available. Additional management strategies like chiller optimisation, enthalpy control, maximum demand, etc., programmes should be freely programmable.
- G. The system controller should have printer connectivity for directly printing critical alarms.
- H. In addition to exchanging data with the management station and the other System Controllers in the same network, a further capability of the system controller shall be to transfer non-critical, global data between DDC/ASC controllers in different groups (i.e. on different buses).
- I. Each System Controller shall operate as a stand-alone controller capable of performing its specified responsibilities independently of other System Controllers in the network.
- J. The System Controller shall be listed with Underwriters Laboratories as compliant with Standard UL916 - Energy Management Systems.
- K. The System Controller shall be designed to be mounted in a mechanical room environment with a rated operating temperature range of 0°C to 50°C.
- L. Each System Controller shall be supplied minimum 32-bit processor with sufficient RAM memory to perform the specified functions. The

memory shall be battery backed up. The battery shall have 8 year normal service life

- M. For system reliability and maintainability the System Controller shall not require a hard disk.
- N. For system reliability each System Controller shall support four independent Local Control Layer Communication Networks. The DDC Controllers, and Protocol Translators shall be divided between the four independent Local Control Layer Communication Networks connected to the System Controller. The Local Control Layer Communication Networks are considered independent if communication traffic on one network does not impact the other network. As such, two networks T-tapped or daisy chained at the System Controller are not considered to be independent. The intent is to split the Local Control Layer Communications Network into four independent networks for fault tolerance and to distribute network communication traffic. The intent is that a wire-to-wire short of the field level communications network not cause loss of communications with all of the Controllers.
- O. The System controller shall have onboard Ethernet card.
- P. The System Controller shall be provided with three serial full duplex EIA-232 ports for the connection of serial data exchange devices or serial alarm printer
- Q. It is the intention to allow a technician to perform advanced monitoring and control functions for any point in the system at any System Controller using a laptop computer. When connected to a System Controller using a notebook computer running the Workstation software, the technician shall be able to access and control any point in the network, limited only by their password definition. The user interface and functionality shall be identical as that provided at a Workstation, except that laptops connected to System Controllers cannot be defined as a report destination.
- R. It is the intention to distribute the database and software features to multiple System Controllers so that the system is scaleable; it can be increased in size without performance bottlenecks, and so that the system is fault tolerant; there is no single point of failure. The operating system of the System Controllers shall be designed to work in this type of distributed processing environment.

2.011 SYSTEM CONTROLLER COMMUNICATION NETWORK

- A. The Network, which connects the System Controllers, shall be Ethernet TCP/IP, operating at 10 Megabaud and complying fully with IEEE 802.3 standard and using Ethernet V2.0.
- B. In addition to the above a dual redundant local area network shall be provided between the System Controllers.
- C. Each system connecting to the IP network should use a full or half duplex 100BaseTX Ethernet connection terminated with an RJ45 jack. Use of standards other than those listed in this specification will not be supported.
- D. The data transmission between the different system outstations and the BMS room should use the Transmission Control Protocol / Internet Protocol. The TCP/IP standard protocol would provide the basis for connection-less, best-effort packet delivery service. The IP should include for ICMP control and error message protocol as an integral part.
- E. Ethernet connections adhere to the 802.3u (Fast Ethernet) standard at the physical layer. 100Base-TX as specified by the 802.3u standard has the following characteristics:
 - Description: UTP (unshielded twisted pair)
 - Type: TIA/EIA1-568-A CAT5
 - Max. Transmit: 100 meters
- F. A standard RJ-45 connector should be used to establish the connection.
- G. At the IP network layer packets traversing this network and network hardware should be governed by the following Internet Standards Process' Request For Comments (RFCs) specifications:
 - 1. RFC 768 User Datagram Protocol (UDP)
 - 2. RFC 791 Internet Protocol (IP)
 - 3. RFC 792 Internet Control Message Protocol (ICMP)
 - 4. RFC 826 Address Resolution Protocol (ARP)
 - 5. RFC 903 Reverse Address Resolution Protocol (RARP)
 - 6. RFC 950 Internet Subnetting Procedures
 - 7. RFC 1042 IP over IEEE 802 Networks

8. RFC 1071 Computing the Internet Checksum

2.012 THE MANAGEMENT LEVEL

A. The Management Level shall consist of a distributed network of Operator Workstations.

B. The function of the Operator Workstation is to provide a user-friendly operator interface and to provide data archiving facilities. For system reliability, the Workstations shall not be required to perform any of the control features.

C. Any PC, which is required to perform the control features as specified, is considered to be a file server. To distribute the processing and for fault tolerance, it is the intention to avoid the use of a file server in the architecture.

D. The Server Computer (To be used as a guide requirement)

The system server computer shall comprise of the following minimum hardware:

(Indicated characteristics shall be used as guiding values as dependant on the development of the industry at the time of quotation and shall serve as a basis for quoting and quote comparisons)

1. The latest available Pentium Processor

2. 1 GB of RAM.

3. 19 "Super VGA monitor.

4. A Super VGA graphics card capable of 1024x768 pixel resolution and 65K colours, non-interlaced (70 Hz or better vertical refresh rate) not less than 4MB of video RAM.

5. A 12 function-key keyboard

6. Mouse pointing device.

7. A Hard disk drive with sufficient capacity (380 GB minimum).

8. A backup unit.

9. A 100 base-T Adapter for Ethernet Networking compatible with TCP/IP network protocols.

2.013 OPERATOR WORKSTATION

The system shall be capable of supporting up to 40 simultaneous Operator Workstation connections using a TCP/IP Local Area Network (LAN) subject to hardware capacity on the server computer. The

Network connection must allow a limitless number of casual users access to the 40 connections on a first-come-first-served basis.

The Operator Workstation shall comprise the following hardware:

(Indicated characteristics shall be used as guiding values as dependant on the development of the industry at the time of quotation and shall serve as a basis for quoting and quote comparisons)

1. The latest available Pentium Processor
 2. 1 GB MB RAM
 3. 19 "Super VGA monitor.
 4. A Super VGA graphics card capable of 1024x768 pixel resolution and 65K colours, non-interlaced (70 Hz or better vertical refresh rate).
 5. Multimedia features.
 6. A Hard disk drive of sufficient capacity (380 GB minimum)
 7. A mouse pointing device
 8. An Adapter for Ethernet Networking compatible with TCP/IP network protocols
- A. Each Workstation shall be provided with one laser jet graphic printer with sufficient memory for graphic reports. Also shall be provided a Colour Ink Jet printer to be used as alarm printer and to print operator-requested information.
- B. Environmental Rating: The workstation, file server and all subsystems shall be capable of operating in an environment of 18 to 28°C with 20 to 80% non relative humidity.

2.014 PORTABLE OPERATOR TERMINAL

- A. The Subcontractor shall provide a portable operator terminal. This terminal shall be provided for command entry, information management, network alarm management, and database management functions.
- B. The laptop terminal shall limit operator access by passwords. The passwords shall be identical to those used in the Operators Workstation.
- C. Operators shall be able to program on-line to any controller using the Portable Operator's Terminal.

- D. The laptop software shall include built-in menus for viewing points by controller, editing and viewing programs, configuring controllers, or communicating to other controllers on the network.
- E. The portable terminal is to be of the latest available model at the time of delivery (Indicated characteristics shall be used as guiding values as dependant on the development of the industry at the time of quotation and shall serve as a basis for quoting and quote comparisons)

Personal Laptop Computer Manufacturer – IBM, HP, Dell, or Toshiba.

- 1. 1 GB RAM (minimum) – Windows XP Professional.
- 2. Latest available Pentium Microprocessor.
- 3. PnP compliant motherboard.
- 4. 250 GB Hard Drive.
- 5. DVD Writer.
- 6. Serial, Parallel Ports and USB
- 7. 1 Keyboard with 83 keys (minimum).
- 8. Integral 2 button Track Point Track Ball or Touch pad.
- 9. 19" SVGA 1026x768 resolution active matrix display.
- 10. Two PCMCIA Type II/III card slot.
- 11. Built in: High Speed Modem. Bluetooth equipped
- 12. WiFi ready - An Adapter for Ethernet Networking compatible with TCP/IP network protocols
- 13. Windows XP Professional
- 14. Memory management software providing a minimum of 600K available base memory and EMS and XMS management services.
- 15. Complete operator workstation software package, including any hardware or software.
- 16. Original printed manuals for all software and peripherals.
- 17. Original installation CD/DVD for all software, device drivers, and peripherals.
- 18. Software registration cards for all included software shall be provided to the Employer.
- 19. Carrying case.
- 20. Spare battery.
- 21. External power supply/battery charger.

F. Proprietary Portable Terminal

1. Manufacturers providing proprietary portable terminals shall submit technical data sheets for the terminal and all associated software and hardware.
2. The proprietary terminal shall meet the same operator interface software requirements as specified above.

2.015 SYSTEM SOFTWARE

- A. The BMS system server shall be based around the Microsoft 64bit multi-tasking environment. Microsoft latest Series software for servers shall be considered.
- B. Standard services supported by the server computer operating system will include the following:
1. Multi-tasking Multi-user support
 2. TCP/IP Network Support
 3. Graphic Display Building Editor
 4. Application software

2.016 THE APPLICATION SOFTWARE

- A. BMS application software shall be configured either locally through a laptop computer or through a workstation connected to the LAN. Application software shall be RAM resident. Application software shall only be limited by the amount of RAM memory. There shall be no restrictions placed on the type of application programs in the system. Each station shall be capable of parallel processing, executing all control programs simultaneously. Any program may affect the operation of any other program. Each program shall have the full access of all I/O facilities of the processor. This execution of control function shall not be interrupted due to normal user communications including; interrogation, program entry, printout of the program for storage, etc.
- B. User Programming Language:
The application software shall be user programmable. This includes all strategies, sequences of operation, control algorithms, parameters, and set points. The source program shall be English language and programmable by the user. The language shall be structured to allow for the easy configuration of control programs, schedules, alarms, reports, telecommunications, local displays, mathematical calculations, passwords, and histories. The language shall be self-documenting. Users shall be able to place comments anywhere in the body of a

program. Program listings shall be configurable by the user in logical groupings.

2.017 LIVE VIDEO

- A. Both the Graphic Display Building Editor and the Operator Interface shall have built in support for the creation and display of live video objects without the need for programming. The Microsoft Multimedia Control Interface (MCI) standard shall be used to facilitate support for a video overlay card. The video overlay card shall enable live video to be fully integrated into custom schematics. The size and position of the video object shall be configured on a per display basis.
- B. Systems, which show the live video, object in a separate window from the operator interface, or on a separate monitor screen shall not be acceptable.

2.018 ENGINEERING TOOLS

- A. Graphic Display Building Editor
 - 1. It shall be possible to launch the Graphic Display Building editor from the operator interface on both the server computer and connected Operator Workstations. It shall allow one step online building of display static and dynamic objects. It shall be a WYSIWYG editor (what you see is what you get) allowing the displays drawn using the editor to appear exactly the same when viewed from an Operator Workstation.
 - 2. Static objects created using the Graphic Display Building Editor shall include static text, rectangles, arcs and circles. However, it shall be possible to animate static objects to give the dynamic characteristics of the real-world object the point represents.
 - 3. It shall be possible to link dynamic objects to the BMS database. They shall allow information to be displayed from the database or to allow an operator to interact with them in order to make changes in the database and to perform control actions. Dynamic objects shall include dynamic text, push buttons, indicators, charts, check boxes, combo boxes and scroll bars.
 - 4. It shall be possible to include static and dynamic display objects on the one display. The editor shall allow display objects to be manipulated by pointing, clicking and dragging. The editor shall allow

display objects to be drawn, re-sized, copied, grouped, rotated, aligned and layered over each other. It shall be possible to copy and paste objects within and between displays.

5. The Graphic Display Building Editor shall support the following features:
 - One step display building (both background and dynamic information)
 - Point and click operation
 - Paste to and from the Clipboard
 - World co-ordinate object placement
 - Ruler and grid
 - Tool, colour and line palettes
 - Dialog boxes for definition of object details
 - Shape and page building
 - On-line help
 - Import graphics from third party packages in the form of Windows Meta files, Bitmaps, TGA and JPEG formats.
 - Standard library of BMS industry objects
 - Three dimensional effects including raised, lowered, ridge and groove line styles
 - Live video element
 - Multi-media element to allow links to web sites and ActiveX Documents to be incorporated onto displays
 - Display Scripts

2.019 APPLICATION PROGRAMMING INTERFACE

- A. Two types of application programming interface (API) are required, the first is for applications written on the BMS server and the second is for applications that are required to run on network based clients (that are not necessarily Operator Workstations).
- B. The BMS system API's must have support for either Visual Basic or C++ or both. Proprietary programming languages shall not be acceptable.
- C. The API on the BMS server requires the following functions as a minimum:
 1. Read and write to points in the database
 2. Access to historical data
 3. Initiate supervisory control actions
 4. Access to the alarm/event subsystem
 5. Access to user-defined database

6. Provide a prompt for operator input.
- D. The API on the network-based clients requires the following functions as a minimum:
1. Read and write to points in the database
 2. Access to historical data
 3. Initiate supervisory control actions
 4. Access to user-defined database
- E. Control Software:
Controllers shall have the ability to perform the following pre-tested control algorithms:
1. Proportional, Integral plus Derivative Control (PID)
 2. Self Tuning PID
 3. Two Position Control
 4. Digital Filter
 5. Ratio Calculator
 6. Equipment Cycling Protection
- F. Mathematical Functions:
Each controller shall be capable of performing Basic mathematical functions (+,-,X,/), squares, square roots, exponential, logarithms, Boolean logic statements, or combinations of both. The controllers shall be capable of performing complex logical statements including operators such as >, <, =, and, or, exclusive or, etc. These must be able to be used in the same equations with the mathematical operators and nested up to five parenthesis deep.
- G. Energy Management Applications:
Controller Panels shall have the ability to perform any or all of the following energy management routines:

1. Time of Day Scheduling
2. Calendar based Scheduling
3. Holiday Scheduling
4. Temporary Schedule Overrides
5. Optimal Start
6. Optimal Stop
7. Night Setback Control
8. Enthalpy Switchover (Economizer)
9. Peak Demand Limiting
10. Temperature Compensated Duty Cycling
11. Airflow Tracking
12. Hot/Cold Deck Reset
13. Free Cooling
14. Hot Water Reset
15. Chilled Water Reset
16. Chiller Sequencing

H. History Logging:

Each controller shall be capable of logging any system variable over user defined time intervals ranging from 1 second to 1440 minutes. Any system variables (inputs, outputs, math calculations, flags, etc.) can be logged in history. A maximum of 32767 values can be stored in each log. Each log shall record either the instantaneous, average, minimum or maximum value of the point. Logs can be automatic or manual. Logged data shall be uploadable to the Operator Workstation for long term archiving based upon user-defined time intervals, or manual command.

2.020 WORKSTATION SOFTWARE

A. Software Description

1. Workstation functions shall include monitoring and programming of all DDC controllers.
2. Monitoring consists of alarming, reporting, graphic displays, long term data storage, automatic data collection, and operator-initiated control actions such as schedule and set point adjustments.
3. Programming of controllers can be done either off-line or on-line from any operator workstation.
4. The software shall be oriented towards operators and programmers. In the operator's mode, all information shall be available in graphic or text displays.

5. Graphic displays shall feature animation effects to enhance the presentation of the data, to alert operators of problems, and to facilitate location of information throughout the DDC system.
6. All operator functions shall be selectable through a mouse. A "windows" environment shall be used to allow multiple functions (windows) to be displayed on the screen simultaneously.

B. System Data Base

The workstation data Base utilizes Microsoft's SQL Data Base Server software, and shall contain all point configurations and programs in each of the controllers that have been assigned to the network. In addition, the data Base shall contain all workstation files including graphic slides, alarm reports, text reports, historical data logs, schedules, and polling records.

C. Object Tree

It shall possible for an operator to view the entire data base through a graphical object tree display. This tree shall present all controllers and their associated points, programs, graphics, alarms, and reports in an easy to understand structure. All object names shall be alpha numeric and up to 16 characters in length.

D. System Configuration

Configuration of the data base shall be through application modules, each having a unique "icon" for easy visual identification. Each module shall provide a windowed menu in which to enter the required data base information.

E. Colour Graphic Displays

The system shall allow for the creation of user defined, colour graphic displays for the viewing of mechanical and electrical systems, or building schematics. These graphics shall contain point information from the data base including any attributes associated with the point (engineering units, etc.). In addition operators shall be able to command equipment or change set points from a graphic through the use of the mouse. Requirements of the colour graphic subsystem include:

1. SVGA, bit-mapped displays. The user shall have the ability to import as background displays.
2. An online graphics drawing editor that provides for all standard geometric shapes, multiple line thickness, shading, up to 32 bit colours, cutting and pasting of objects, inclusion of text, and zooming.

3. Built-in control panel objects such as buttons, knobs, gauges, line graphs, etc. to enable operators to interact with the graphic displays in a manner that mimics their mechanical equivalents found on field installed control panels. Using the mouse, operators shall be able to adjust set points, start or stop equipment, modify PID loop parameters, or change schedules.
4. Status changes or alarm conditions can be highlighted by objects changing screen location, size, colour, text, blinking or changing from one display to another.
5. Ability to link graphic displays through user defined objects, alarm testing, or the result of a mathematical expression. Operators shall be able to move from one graphic to another by selecting an object with a mouse, no menus shall be required.

2.021 MONITORING AND CONTROL

A. Automatic Monitoring

The software shall allow for the collection of data and reports from any Controller, RSCU or SDCU controller through either a hardwire or modem communication link. The time schedules and content of the polling shall be user configurable and include any subset of the controller's data base including application programs.

B. The system shall support acquisition of data using the following techniques:

- a. Periodic Scanning
- b. Report by Exception

C. The system shall support a range of scan intervals, ranging from less than 1 second up to 15 minutes.

D. In order to minimise communications traffic, the system shall automatically block data requests using contiguous addresses and the scan interval to generate scan packets, optimising throughput for a given scanning load. The system shall also provide utilities to examine scan packet allocation for each scan interval, and compile aggregate statistics on communication link usage.

E. Where supported by the controlling device, Report by Exception (RBE) protocols shall be used to reduce the scanning load of the system while improving system response. If necessary, periodic scanning may be used in conjunction with RBE to ensure data integrity.

2.022 SYSTEM DATABASE

- A. The system shall provide a real-time database incorporating data from analogue, logical or pulse inputs. The database shall be configurable by the end user without the need for any programming and shall be able to be modified on-line without interrupting operation of the system. In addition to point based information, the database shall also provide historization capabilities for analogue, digital, pulse and event based information. This information shall be accessible by all facilities of the system such as custom displays, reports, trends, user written applications, etc.

2.023 DATABASE STRUCTURE

- A. The real-time database shall support collection of data and storage using the following structures:
 - 1. Access Point Structures
 - 2. Analogue Point Structures
 - a. Status Point Structures
 - b. Accumulator Point Structures
 - c. Historical Data Structures
 - d. Event Data Structures
 - e. User Defined Structures
- B. Each of the Point database structures shall be comprised as a composite point with a number of associated parameters which may be referenced relative to a single tagname. Specifically, each of these parameters shall be accessible by various sub-systems such as the Graphical Operator Interface, Report Generation system and Application Program Interface in a simple POINT.PARAMETER format without the need to know any internal storage mechanism.
- C. The system shall maintain portions of the database requiring frequent high-speed access as memory resident information and other less frequently accessed data as disk resident data. Memory resident data shall be checkpoint to disk every minute to minimise loss of data in the event of loss of power or other system failure.
- D. Database backup shall be possible with the system on-line including backup of historical based data. The backup shall be possible via Microsoft Windows operating system utilities (latest OS software version shall be provided).

- E. Point data shall be stored in a composite point database structure that provides a wide range of configurable information including but not limited to:
 - 1. Point name and description
 - 2. Multiple locations for data storage and device scanning addresses
 - 3. Scan period
 - 4. Multiple types and instances of alarms
 - 5. Multiple dead band or hysteresis settings (analogue points)
 - 6. Monitoring and Control access restriction information
 - 7. Location of operator alarm handling instructions
 - 8. Location of ancillary information associated with the point.
- F. Analogue Points
Analogue data
- G. Status Points
Status (digital) information shall be stored in a Status Point type in the database. The status point shall be a composite point capable of processing from a single to a three-bit digital input, allowing up to eight possible states.
- H. Accumulator Points
Data associated with pulsed inputs shall be stored in the system in a composite point database structure that shall provide automatic tracking of instrument rollover.
- I. Grouping of Points
The BMS system shall provide a means by which a number of alarm inputs, outputs and other related points can be grouped together for more convenient monitoring and control without the need for custom graphics.

2.024 HISTORY MANAGEMENT

- A. Collection of historical point data shall be configurable as part of the point definition. Once configured, this data shall be collected automatically. Historical data collection shall be provided for both snapshots and averages with intervals ranging from 5 seconds to 24 hours.
- B. Once assigned to history, point data shall be available by POINT.PARAMETER access used in conjunction with a history offset to locate the particular value of interest. The graphical operator interface,

trend, report generation and application interfaces shall be able to access historical data.

- C. Modifications to the history collection of a point shall be possible on-line without the loss of previously collected data for the point being changed or any other points in the system currently being historized.

2.025 TRENDING

- A. The system shall provide flexible trending allowing real-time, historical or archived data to be trended in a variety of formats. In addition, trend data types shall be able to be combined to allow for comparisons between data e.g. current real-time data versus archived data.
- B. Trend Capabilities
The system shall provide trending capability with the following functions:
 - 1. Real time trending
 - a. Historical trending
 - b. Archived History trending
 - c. Trend Scrolling
 - d. Trend Zoom
 - e. Engineering Unit or Percent
 - f. Cursor readout of trend data
- C. Provide the Trend comparisons between archived, real-time and historical data (for example, this year vs. last year). Comparisons between the same point offset in time, or different points must be possible.
- D. Trend Decluttering via per-pen enable/disable on multi-plot style trends
- E. Independent Y-axis per point on multi-plot style trends. It must be possible to display the Y-axis for any point on the trend by simply selecting the point using the mouse or keyboard
- F. copying the currently displayed trend data to the clipboard for pasting into spreadsheet or document

- G. Configuration of trends shall only require the entry of the Point Name into the desired trend template to produce the trend. All trend configuration must be possible on-line without interruption to the system. Historization of data shall not be affected by changes to trend configuration. Systems that only provide trending via a third party package will not be acceptable.
- H. Trend Types: The system shall be able to present real-time, historical or archived data in a variety of formats, including single, dual and multiple value trends of up to 8 points. For each trend set display it shall be possible for operators to configure the number of historical samples and ranges displayed. Points configured in trend sets shall be changeable on-line.
- I. Operators shall be able to zoom in on information displayed on trend sets for closer inspection by dragging out an area of interest with the mouse or other pointing device. From such a selection, it shall be possible to copy the underlying data to the Windows clipboard for subsequent pasting into a spreadsheet application such as Microsoft Excel. Scroll bars shall be available to move the Trend set backwards and forwards across the historical records. The trend sets shall automatically access archived history files without operator configuration.
- J. It shall be possible to embed trend objects as part of custom displays. The following formats shall be available:
 - 1. Bar Trend
 - 2. Line Trend
 - 3. Numeric Trend
 - 4. Tuning Trend
 - 5. Pie Trend
 - 6. X-Y Plot

2.026 EVENT MANAGEMENT

- A. It shall be possible to log an event such that it shall be journalized in the event file and optionally printed on the event printer. The journal shall contain the following event information:
 - 1. Alarms
 - 2. Alarm Acknowledgements
 - 3. Return to Normal
 - 4. Operator Control Actions
 - 5. Operator Login & Security Level Changes
 - 6. On-line database Modifications

7. Communications Alarms
 8. System Restart Messages
- B. Standard Displays shall be provided to show the current journal file with the most recent event at the top of the display. Subsequent page forward actions shall allow display of progressively older events. Sorting and filtering of the journal shall be possible via a standard report, which shall be configurable by filling-in-the-blanks. Coding or scripting of any kind shall not be required. If events are viewed on the screen, it shall be possible to pause the real time view and prevent events from scrolling off the screen as new events occur.
- C. The Event database entries shall contain the following information:
1. Time & Date Stamp
 - a. Point Name
 - b. Event Type
 - c. Alarm Priority
 - d. Point Description
 - e. Point value or status at time of the event
 - f. Engineering Units
- D. The Event database must also be accessible from other sub-systems such as the Operator Interface, Report Generation and Application Programmers Interface.
- E. It shall be possible to have an on-line event file as large as the disk capacity can accommodate. For example, given the appropriate disk space it shall be capable of storing up to 1,000,000 (one million) events on-line.
- F. The event file shall store events in a current online buffer. When the buffer is full an alarm shall be raised advising the operator to save the file to an external media. The current online buffer contents shall then be transferred to an archive buffer to await archiving to an external media. The current online buffer shall, without interruption, continue to store current events.
- G. Another file area shall be available to hold archived event files ready for playback. These are event files previously archived to external media. Operators shall be able to restore previously archived files via the operator interface and a dedicated display.

- H. The events file system shall be fully integrated with the standard reporting system. The system shall automatically reference the restored playback file if a Report is requested containing a time search window covered by the current playback file.
- I. The operator shall be able to restore previously archived files and review or print them from the Operator Workstations.

2.027 ALARM MANAGEMENT

- A. The BMS shall support several different types of alarm, including:
 - 1. Two high-value alarms
 - a. Two low-value alarms
 - b. Two deviation alarms
 - c. Rate of change alarm
 - d. Unreasonable value alarm
- B. Any four of these alarms shall be assignable to each Analogue or Accumulator point on an individual point basis as part of the point configuration process.
- C. Alarm Priorities: Each monitored point in the system shall be able to assigned one of four alarm priorities to individual states. The meaning of the priorities shall be as follows:
 - 1. Journal: Changes of state shall be journalised to the Alarm/Event Log and optionally printed on the Alarm/Event printer.
 - 2. Low: Change of state will generate a Low priority alarm, which will appear on the Alarm Summary. Optionally, the alarm may be printed on the Alarm/Event printer or generate an audible tone.
 - 3. High: Change of state will generate a High priority alarm, which will appear on the Alarm Summary. Optionally, the alarm may be printed on the Alarm/Event printer or generate an audible tone.
 - 4. Urgent: This is the highest priority. Change of state will generate an Urgent priority alarm, which will appear on the Alarm Summary. Optionally, the alarm may be printed on the Alarm/Event printer or generate an audible tone.
- D. Within each of the four alarm types there shall be 15 sub-priorities available.

- E. It shall be possible to configure a time such that if a low priority alarm is not acknowledged within this time the alarm's priority is elevated to high priority. If a high priority alarm is not acknowledged within a configured time, its priority is elevated to urgent priority.
- F. It shall be possible to associate additional messages to be logged into a message summary in the event of an alarm condition.
- G. When an alarm is acknowledged, it shall be possible to automatically issue a reset to a controller to indicate the alarm is acknowledged and to attempt to reset the alarm point.
- H. Alarm Enunciation: Alarms shall be enunciated by:
 - 1. Most recent, highest priority alarm message appearing on dedicated alarm banner on the operator interface.
 - 2. Alarm message appearing on alarm summary display.
 - 3. Available Tone - based on a "*.wav" or other sound file for each alarm priority
 - 4. Alarm message printed on the alarm printer
 - 5. Alarm indicator flashing on the operator interface
- I. Alarms shall be enunciated at the Operator Workstation even if there is no operator currently signed-on. This feature shall be available on network connected Operator Workstations as long as the computer running the Operator Workstation software remains logically connected to the network. If the Operator Workstation is minimised in the Windows environment, then the Operator Workstation icon will indicate an alarm. An audible tone shall be able to be generated and this tone shall be specified by a "*.wav" or other sound file for each alarm priority.
- J. Points shall be enunciated whilst in alarm. If a point is set to alarm inhibited the point shall no longer cause annunciation. If a point goes into an alarm state whilst inhibited and then is still in the alarm state when the point is set to alarm enabled, the point shall immediately cause annunciation.

- K. Alarm Processing : Assigning an alarm to the point shall automatically cause the system to perform the following actions when an alarm occurs:
1. The alarm shall be time stamped to the nearest second and logged in the Event database with the Point Name, Alarm type, Alarm Priority, Point Description, New value and Engineering Units
 2. The point value which is in alarm shall turn red and flash on any standard or custom display which uses that point
 3. An Unacknowledged alarm entry shall be made in the system alarm summary for Low, High and Urgent Alarms
 4. The audible alarm shall sound (if configured)
 5. The alarm annunciation indicator shall flash
- L. In addition, the alarm banner of the Operator Interface must show the most recent (or optionally oldest), highest priority, unacknowledged alarm in the system.
- M. Dedicated Alarm Banner and Alarm Indicator: A dedicated alarm banner shall appear on all displays showing either the most recent or oldest (configurable), highest priority, unacknowledged alarm in the system. This banner shall be clear when there are no unacknowledged alarms for the operator to process.
- N. On occurrence of an alarm, the graphic display shall output the point identification, point type, and description on the banner. If multiple alarm/change of state conditions occur, subsequent messages shall overwrite the display if they are higher priority. As subsequent alarms are displayed, the previous alarm information shall move to an unacknowledged alarm list, awaiting acknowledgement by the operator.
- O. An alarm indicator shall also appear on all displays. This indicator will flash red when there are any unacknowledged alarms pending in the system. This indicator will remain solid red if there are alarms, which have not returned to normal but which have all been acknowledged. The indicator will be clear if there are no points in an alarm condition.
- P. Alarm Logging
As well as being logged on the printer, alarms shall be logged to an event file for future retrieval in alarm reports or archived to removable media.

2.028 ALARM RESPONSE FUNCTION KEYS

- A. The following dedicated function keys shall be provided on the keyboard for alarm action:
 - 1. ACKNOWLEDGE: After moving the cursor to the point in alarm on the screen and selecting the point the operator shall be able to acknowledge an alarm by pressing this key. This action shall be logged in the event file and on the printer showing the operator ID with the message.
 - 2. ALARM SUMMARY: By pressing a dedicated key at any time the operator shall be able to view a display showing all currently active alarms. The alarm messages shall be colour-coded showing priorities. The operator shall be able to view the alarms according to priority. It shall be possible to acknowledge alarms from this display and also go to the associated display defined for the point.
 - 3. ASSOCIATED DISPLAY: After moving the cursor to the point in alarm on the screen and selecting the point the operator shall be able to bring up the display applicable to that alarm by pressing this key. Just selecting the associated display key directly will bring up the associated display for the point currently on the alarm banner.
- B. Alarm Acknowledgement : The system shall provide for efficient alarm acknowledgement in a number of ways as follows:
 - 1. Selection of any POINT.PARAMETER from a custom graphic and pressing the dedicated acknowledge push-button
 - 2. Selection of the alarm banner and pressing the dedicated acknowledge button
 - 3. Selection of the alarm in the alarm summary display and pressing the dedicated acknowledge button
 - 4. By performing a page acknowledge from the alarm summary display
- C. On acknowledgement by the operator, the flashing indicator shall turn steady, and the point value shall remain red on any system or custom graphic. The acknowledgement shall also be logged in the Event database identifying the operator or station that acknowledged the alarm. If the point goes out of alarm before being acknowledged by the operator, the alarm shall be shown by a different indication and remain in the list until specifically acknowledged by the operator.
- D. Alarm Filtering: The Alarm Summary shall be able to filter the alarms displayed to the operator. The filtering criteria shall include as a minimum:
 - 1. Individual Priorities (i.e. Urgent, High, Low)

2. Ranked Priorities (i.e. Urgent only, Urgent & High only, Urgent, High & Low)
 3. Unacknowledged Alarms only
 4. Individual Areas only
- E. Additional Alarm Information: The BMS system shall provide support for an additional message to be tagged to the alarm. This message shall provide the operator with additional information on the alarm but shall not clutter the alarm summary. It shall appear in a separate message summary at the same time as the alarm appears in the alarm summary. The messages can be pre-configured and then simply attached to individual points by means of a message ID.

2.029 ADVANCED ALARM MANAGEMENT

- A. The BMS shall be capable of advanced alarm management, which includes set stages of alarm handling.
- B. The stages shall be:
1. Silence alarm condition
 2. Acknowledge and action alarm condition
 3. Respond to alarm condition by using pre-defined responses
 4. Optionally reset alarm
- C. All actions shall be recorded in the event file for retrieval and auditing purposes.
- D. When an alarm is silenced, an instruction page for the alarm will be displayed. The alarm may then be acknowledged from this page and actioned.
- E. Once the alarm is acknowledged and appropriate action has been taken, the operator may move to the response page to select from up to 100 user defined responses to be logged in the event file. Alternatively, the operator shall be able to enter their own response, which will also be logged in the event file. At the same time the alarm is removed from the alarm file.
- F. It shall be possible to enable/disable this feature on a point by point basis given the appropriate system privilege level.

2.030 REPORTING

- A. The system shall support a flexible reporting package to allow easy generation of report data. The reports provided shall include pre-

configured standard reports for common requirements such as Alarm Event reports and custom report generation facilities that are configurable by the user.

- B. Standard Reports: The following pre-formatted reports shall be available on the system:
 - 1. Alarm/Event Report
 - 2. Operator Trail Report
 - 3. Point Trail Report
 - 4. Alarm Duration Report
 - 5. All Point Report
 - 6. After Hours Alarm Report
 - 7. Point Attribute Report
- C. Configuration of these reports shall only require entry of the schedule information, and other parameters such as Point Name or wildcard, Filter information, time interval for search and destination printer to fully configure the report. Specifically, no programming or scripting shall be required.
- D. Alarm/Event Report :A report shall be provided to produce a summary of all events of a specified type for nominated points occurring in a time period. The time period may be specified as an absolute start and end date and time, or as a period relative to the current time. This report shall also be able to produce a summary of all changes made by a specific operator.
- E. Operator Trail Report: A report shall be provided to produce a summary of all operator actions relating to a specific operator in a specified period.
- F. Point Trail Report: A report shall be provided to produce a summary of all events of a specified type occurring in a period on nominated points.
- G. Alarm Duration Report: A report shall be provided which calculates the total amount of time a nominated point or group of points has been in an alarm condition over a given time period. The time period may be specified as an absolute start and end date and time, or as a period relative to the current time.
- H. All Point Report: A report shall be provided to produce a list of point information, including point name, description, point type, engineering

units, and current values. Report configuration shall allow filtering based on a wide variety of criteria.

- I. After Hours Alarms Report: A report shall be provided to produce of summary of all Alarms occurring during the period specified by the operator as "After Hours"
- J. Point Attribute Report: A report shall be provided to points selected by one of the following attribute criteria:
 - 1. Out-of-service
 - 2. Alarm suppressed
 - 3. Abnormal input levels
 - 4. In Manual mode.
- K. Additional Custom Reports: In addition, configurable report generation facilities must be provided to allow custom reports to be produced. They shall be able to be configured at any time with the system online, and shall be able to access any database values. At least two methods of custom report generation shall be available, including the following:
- L. Microsoft Excel
The BMS shall provide the facility for the use of Microsoft Excel as a reporting tool – allowing calculations such as summations, maxima, minima and standard deviations, and the production of graphs, charts and tables.

Data accessible for Excel reporting shall include alarms, events, and point parameter values.

2.031 ODBC

- A. The BMS shall be capable of providing selected data in an ODBC format for the purpose of extracting data and creating custom reports. It shall be possible to access tables of data from the BMS through an ODBC compliant tool such as Crystal Reports.
- B. It shall be possible to incorporate the activation of custom reports created through the ODBC compliant tool through the standard BMS report subsystem. Example reports shall be provided to illustrate how to access the ODBC data in the BMS.
- C. A report detail display shall allow naming of reports, scheduling information and the destination of the report. The report destination shall be a printer, operator interface or internal file. The report output format

shall be HTML (Hypertext Mark-up Language), Microsoft Word or RTF format.

2.032 REPORT ACTIVATION

- A. Reports shall be activated in one or more of the following ways:
 - 1. Periodic activation at user specified intervals
 - 2. Operator Demanded
 - 3. Event Initiated e.g. Change in point value
 - 4. Application Initiated

2.033 USER DEFINABLE DATABASE

- A. In order to support other types of data such as user entered or calculated data from application programs, the system shall also provide a User Definable database area that can be fully integrated into the system. Data contained in this database must be accessible by:
 - 1. Custom Graphics
 - 2. Custom Reports
 - 3. Application Programs
 - 4. Network Applications using a Network API

2.034 POINT INITIATED PROGRAMS (PIPS)

- A. In addition to standard point processing functions, the system shall allow additional processing through the use of standard PIPS that may be attached to an analogue, status or accumulator point. Typical functions to be provided by these PIPS are listed below:
 - 1. Arithmetic Calculation
 - 2. Boolean Calculation
 - 3. Maximum/Minimum Value
 - 4. Composite Alarms
 - 5. Integration
 - 6. Run Hours Totalisation
 - 7. Group Alarm Inhibit
 - 8. Report Request by Point Change
 - 9. Application Program Request (By point value change or cyclic period)
 - 10. Alarm Transportation
 - 11. Value Transportation
 - 12. Door Activity Task Request

- 13. Security Area Seal / Unseal
- 14. Alarm or Point Value Change Graphic Call-up
- 15. Value Change Group or Area Alarm Inhibit

2.035 HISTORICAL DATA ARCHIVING

- A. The system shall support archiving of historical data to allow a continuous record of history to be built up over a period of time. Archived data may be stored on the hard disk of the system or a remote network drive or moved off-line to removable media such as floppy disk, cartridge tape, DAT tape, or optical disk. The number of archives maintained on the system before transferral to off-line media shall only be limited by the size of the hard disk or remote network drive. The system shall allow the user to define the specific intervals of history to be archived to avoid archiving of unnecessary data.
- B. Archiving of historical data may be activated by one of the following methods:
 - 1. Operator Demand
 - 2. Periodic Schedule
 - 3. Event Initiated
- C. Once archived, the data shall be available for re-trending through the system trend facilities in combination with the current on-line history or other archives. Providing the archived history is present on the BMS Server's hard disk or remote network drive, the trend facilities must be able to access it transparently for display, when a user scrolls beyond current on-line history limits.

2.036 TIME SCHEDULES

- A. A minimum of one thousand (1000) BMS based time schedules shall be provided by the system.
- B. The time schedule facility shall allow the scheduling control of points on both a periodic and one-off Basis by the BMS. The time schedule facility shall execute every 60 seconds and it shall be possible to schedule Point controls on minute boundaries.
- C. All time schedules shall be configurable via the Operator Workstation.
- D. Each time schedule entry shall consist of:
 - 1. Date
 - 2. Time
 - 3. Point name

4. Point parameter
 5. Target value
 6. Type of scheduling
- E. The available time schedule type shall be as a minimum:
1. One shot - to be executed only once then deleted
 2. Daily - to be executed every day
 3. Workday - to be executed Monday to Friday
 4. Weekend - to be executed on Saturday and Sunday
 5. Holiday - to be executed on holidays
 6. Individual days - to be executed on individual days (eg. Monday)
- F. Where the control device supports an internal time schedule program, the BMS shall be able to upload, display, modify and download the control device time schedules. Support for the control device time schedules shall be in addition to the BMS time schedules.
- G. Programmer's Environment
- The programmer's environment shall include access to a superset of the same programming language supported in the Controller or RSCU controllers. Here the programmer shall be able to configure application software off-line (if desired) for custom program development, write global control programs, system reports, wide area networking data collection routines, and custom alarm management software.
- H. Security
- The software shall employ a two tiered password system. The first tier shall consist of the user's name. The second tier shall be a unique password consisting of 8 alphanumeric characters. Each password shall have a unique access level. At least 8 levels shall be definable.
1. In response to operator log-on or log-off, the software shall automatically run a program that performs actions such as running a report, generating a specific graphical display, displaying the active alarm list, etc.
 2. In addition, it shall be possible to provide security overrides on any portion of the software by user. For example, this would allow an operator with view only privileges to modify the space set point in the conference room, but not modify any other set points in the system.
 3. There shall be an inactivity timer adjustable in software that automatically logs off the current operator after the timer has expired.

I. Saving/Reloading Programs

The workstation software shall have an application to save and restore field controller memory files. Default values stored in the workstation data base shall be sent to the controller during a reload operation either automatically or at the user's option.

2.037 COMMUNICATIONS AND CONNECTIONS

A. HVAC, Fire Alarm and detection system, Emergency Lighting System and Life Safety System (Security and CCTV system will be included in a separate contract) shall be fully integrated with the Building Management System (BMS), where by all the above system can be monitored and controlled from a common workstation, with one single alarm and monitoring history and alarm/operator data base log of all the integrated systems. Multiple windows showing the various systems on a single screen without a common database will not be acceptable. The integration & compatibility between the various systems, which shall be made into the BMS, shall be confirmed in writing by the return of the tender by each of the individual system suppliers. Further to this the Subcontractor will have to provide reference installation in the Middle East of such a successful integrated solution.

B. The proposed network has been designed to integrate all services within a single network.

C. The BMS system shall provide communications over a variety of physical media topology's as follows:

1. RS-232
2. RS-422
3. Ethernet
4. Proprietary Networks

D. The system shall be capable of supporting greater than ninety separate communications links to networks of control devices. Each connection shall operate independently of the others and facilities shall be provided by system displays to individually place these links in service or out of service.

- E. Where supported by the controller, it shall be possible for serial connections to the BMS Server to be routed via a terminal server and the LAN as an alternate to connecting directly to the host computer. TCP/IP based Terminal Servers are suitable and must be Ethernet connected to the BMS server directly.
- F. Given the sufficient level of system privilege, it shall be possible to view, manipulate and analyse all data in the system from any Operator Workstation in the system, including those operating remotely via dial-up modem links.
- G. Once a control device is configured and placed in service, the system shall automatically begin background diagnostic scanning of the device to ensure that communications are monitored independently of any monitoring scanning.
- H. The system shall perform checks on data integrity of all data acquired from the device. If an invalid or time out response is received, the data shall be ignored and the system will record the transaction as an error. Statistics shall be kept and displayed by the system on errors encountered in communication by means of a communications barometer. The barometer shall increment for every failed call and decrement for each successful call. In addition, the system shall alarm separate marginal and failure conditions based on user defined limits to advise the operator of the device and link that has failed. Communications statistics shall be displayed as standard on the system and shall also be available as part of the reporting system or custom displays.

2.038 HARDWIRED CONNECTIONS

- A. Analogue and digital signal values shall be passed from one system to another via hardwired connections.
- B. There will be one separate physical point on each system for each point to be integrated between the systems.
- C. Analogue points will be 4-20 mA signals originating at the "from system" and being received by the "to system".
- D. Digital points will be "dry contact" signals originating at the "from system" and being received by the "to system."

2.039 OPEN INTEGRATION

- A. Any of the following Open Protocol Standards shall be used for integration of 3rd party devices or systems

2.040 DATA EXCHANGE

- A. The BMS system shall have the capability to interface to the point database of other similar BMS systems (i.e. nodes) on a TCP/IP network. This shall enable both the acquiring of point data and issuing control outputs to other BMS systems.
- B. Data Exchange with Microsoft Excel: The system must be capable of exporting bulk data to Microsoft Excel.
- C. As a minimum the following shall be supported:
 - 1. Allow retrieval of data either periodically or snapshot
 - 2. Allow retrieval of data via POINT.PARAMETER requests
 - 3. Allow retrieval of tag names, descriptions etc
 - 4. Allow retrieval historical data
 - 5. Writing of values from Excel back to the supervisory system

2.041 DIRECT PROTOCOL (INTEGRATOR PANEL)

- A. The BMS system shall include appropriate hardware equipment and software to allow bi-directional data communications between the BMS system and 3rd party manufacturers' control panels. The BMS shall receive, react to, and return information from multiple building systems, including but not limited to the chillers, boilers, variable frequency drives, power monitoring system, lighting control, emergency and exit lighting, CO monitoring system, fire alarm and detection system and stand-by generator supply, .
- B. All data required by the application shall be mapped into the Network Controller's database, and shall be transparent to the operator.
- C. Point inputs and outputs from the third-party controllers shall have real-time interoperability with BMS software features such as: Control Software, Energy Management, Custom Process Programming, Alarm Management, Historical Data and Trend Analysis, Totalization, and Dial-Up and Local Area Network Communications.
- D. The Building Management System shall provide any combination of third-party controllers on a single network. A minimum of 100 third-party controllers shall be supported on a single network. Integration shall be via RS-232 or RS-485 technologies.

- E. The system operator shall have the ability to verify, and diagnose communication messages and point information between third-party controllers and the BMS.

2.042 NEUTRAL PROTOCOL INTEGRATION – BACNET (ANSI / ASHRAE 135 1995)

- A. The neutral protocol used between systems which allows the monitoring of BACnet Servers via data acquisition and control. shall be BACNet over Ethernet and comply with the ASHRAE BACNet standard 135-1995.
- B. The BACNet System Integrator (BSI) will allow bi-directional communications between the host system and a BACNet system over an Ethernet data link. Supported media shall include fiber, 10base2, and 10baseT.
- C. A complete Protocol Implementation Conformance Statement (PICS) shall be provided for the Integrator and all BACNet system devices.
- D. The BSI shall conform to BACNet conformance class 4, and provide the ability to monitor and control BACNet system points from the host system and host system points from the BACNet system.
- E. The ability to share data and change of state (COS) between the host and BACNet systems shall be provided.
- F. BACnet functional groups:
 - 1. Clock
 - 2. COV Event Response
- G. BACnet standard application services:
 - 1. Confirmed COV Notification
 - 2. Subscribe COV
 - 3. Unconfirmed COV Notification
 - 4. Read Property
 - 5. Write Property
 - 6. Who-Is
 - 7. I-Am

H. BACnet standard objects:

1. Analogue Input
2. Analogue Output
3. Analogue Value
4. Binary Input
5. Binary Output
6. Binary Value
7. Device
8. Multi-state Input
9. Multi-state Output

2.043 NEUTRAL PROTOCOL INTEGRATION – INDUSTRIAL STD (MNI)

- A. The neutral protocol used between the BMS network and integrated systems will be an Industry Standard protocol, such as Allen-Bradley Ethernet, Modbus+, or Eurotherm-LIN.
- B. Metasys Network Integrator (MNI) shall link the BMS network with one or more integrated systems via an interface driver supporting one of the above protocols.
- C. MNI shall connect the BMS to multiple integrated systems concurrently, without requiring modification of the existing BMS architecture.
- D. MNI shall be able to link diverse 3rd party components and systems, and shall provide the capability to add and integrate other components.
- E. MNI shall employ peer-to-peer communication between the integrated systems and one or more NCM Network Ports, allowing bi-directional exchanges of communication and data.
- F. MNI shall provide the tools for building, modifying, and updating both point-mapping and generic applications. Through MNI, the BMS shall monitor and control data points related to the integrated systems.
- G. The MNI software and hardware, in conjunction with the integrated systems, shall translate data from multiple systems into one system and one user interface. The MNI shall allow an operator to monitor and control data and systems through an Operator Workstation (OWS).

- H. The ability to share data and change of state (COS) between the BMS network and the integrated systems shall be provided via the MNI interface.

2.044 CLIENT-SERVER (OPC)

- A. The neutral protocol used between the BMS and diverse systems will be the Industry Standard protocol OLE for Process Control (OPC) over Ethernet.
- B. The BMS server shall provide an integrated OPC Client and integrated OPC Server.
- C. The OPC Client shall support the mandatory OPC Release1.0A as a minimum.
- D. The OPC Server shall support the mandatory OPC Release1.0A custom and automation OPC interfaces as a minimum.
- E. OPC shall link the BMS with one or more systems via OPC Servers, providing an industry-standard, component-based interface to BMS data.
- F. The OPC Server shall have the following characteristics and functionality:
 - 1. Run on Windows XP Professional or with latest OS machines connected via Ethernet to the BMS.
 - 2. Allow generic components to be used and shared with other OPC-compliant clients and systems.
 - 3. Serve as the "translation service" for the various languages used by the diverse systems with which it communicates.
 - 4. Support multiple OPC clients running on both local and remote machines in the network.
 - 5. Control the interaction and communication between workstation applications (on clients) and the BMS by providing standard methods for sharing and exchanging data between the BMS and other systems.
 - 6. Provide interfaces for browsing, reading, and writing accessibility.
- G. OPC shall connect the BMS to multiple systems concurrently, without requiring modification of the existing BMS architecture.
- H. OPC shall seamlessly integrate diverse proprietary components and systems, and shall provide the capability to add and integrate other components.
- I. The OPC software shall translate data from multiple systems into one user interface. The software shall allow an operator to monitor and

control data, communication, and systems through an OPC client (workstation).

- J. The ability to share data and change of state (COS) between the BMS network and the integrated systems shall be provided via the OPC interface.

2.045 WEB BROWSER ACCESS TO THE BMS

- A. Web-page controls and a web server interface to the BMS shall be provided, which allow tenants or other users to monitor & control a variety of BMS-supervised functions via a web page viewed from a standard web-browser.
- B. It shall be possible to limit web browser access to BMS facilities by means of standard web and networking techniques.

2.046 SYSTEMS INTEGRATION/BMS SPECIFIC REQUIREMENTS

- A. Chiller Integrator Interface
 - 1. The BMS system shall include appropriate hardware equipment and software to allow two way data communications between the BMS system and the chiller manufacturer's chiller control panel.
 - 2. It shall be the responsibility of the Subcontractor to coordinate with the chiller manufacturer to provide a functional data communications connection.
 - 3. All data supported by the chiller communication protocol shall be mapped into the supervisory DDC controller's database and shall be displayed on a chiller data screen at the Operator Workstation and shall be transparent to the operator.
 - 4. The Subcontractor shall furnish either the OSP or BACnet communications interface as required by the chiller manufacturer.
 - 5. The Subcontractor shall provide all communications and power wiring and gateway panel installation for the DDC system. The chiller manufacturer shall provide all hardware for connection of the manufacturer's processor.
 - 6. The Subcontractor shall provide all hardware and software required for the chiller manufacturer's gateway interface.

B. VFD INTEGRATOR INTERFACE

1. The BMS system shall include appropriate hardware equipment and software to allow two way data communications between the BMS system and the VFD manufacturer's control panel.
2. It shall be the responsibility of the Subcontractor to coordinate with the VFD manufacturer to provide a functional data communications connection.
3. All data supported by the VFD communication protocol shall be mapped into the supervisory DDC controller's database and shall be displayed on data screens at the Operator Workstation and shall be transparent to the operator.
4. The Subcontractor shall furnish either the OSP or BACnet communications interface as required by the VFD manufacturer.
5. The Subcontractor shall provide all communications and power wiring and gateway panel installation for the DDC system. The VFD manufacturer shall provide all hardware for connection of the manufacturer's processor.
6. The Subcontractor shall provide all hardware and software required for the VFD manufacturer's gateway interface.

2.047 DDC SENSORS AND POINT HARDWARE

- A. DDC Sensors and Point Hardware are included within the Section 15900 HVAC Instrumentation and Control.

Part 3 EXECUTION**3.01 HARDWARE INSTALLATION**

- A. As per details from Section 15900 HVAC Instrumentation and Controls
- B. This installation includes for “gateway switches” in each tower located in the telecom room in the central core, for the connection of any devices.
- C. The requirements for implementing phase 1 of the network can be divided into the following areas:
 - 1. BMS security room located in the P2 level of the south podium.
 - 2. Fibre optic backbone cabling between the BMS room linked in a ring configuration between each tower IT room in the P2 level and the central ramp plant rooms.
 - 3. Tower IT equipment room on level P2 in each tower, the riser and floor telecom room at the access point to the riser including the roof plant room.
 - 4. Cables and outlets for connecting any point into the building management system.

3.02 CALIBRATION AND ADJUSTMENTS

- A. After completion of the installation, perform final calibrations and adjustments of the equipment and sensors provided under this contract and supply services identical to the proper performance of the BMS and other integrated systems.

3.03 SOFTWARE INSTALLATION

- A. The Subcontractor shall provide all labour necessary to install, initialize, start-up and debug all system software as described in this section. This includes any operating system software or other third party software necessary for successful operation of the system.
 - 1. Data base Configuration - the Subcontractor shall provide all labour to configure those portions of the data base that are required by the points list and sequence of operation.
 - 2. Colour Graphic Slides - unless otherwise directed by the Client, the Subcontractor shall provide colour graphic displays as depicted in the Contract drawings for each system and floor plan. For each system or floor plan, the display shall contain the associated points identified in the point list and allow for set point changes as required by the Client.

3. Reports - the Subcontractor shall configure a minimum of 6 reports for the Client as listed below:
- Central Plant Status Report
 - Air Handler Status Report
 - Energy Consumption Report
 - Space Temperature Report
 - Specialty Equipment Status Report
 - As built software documentation shall include the following:
 - Descriptive point lists
 - Application program listing
 - Application programs with comments.
 - Printouts of all reports.
 - Alarm list.
 - Printouts of all graphics

3.04 TESTING, BALANCING, COMMISSIONING AND SYSTEM STARTUP:

- A. Commissioning is a mandatory documented performance requirement of the Subcontractor and subcontractors for all control systems under this contract detailed in Specifications drawings and sequence of operations. Testing, Balancing and Commissioning shall be as detailed in standards listed in this Section as well as per Section 15950 Testing, Adjusting and Balancing .
- B. Documentation shall be presented upon completion of each commissioning step and final completion to ensure proper operation of the Building Management System.
- C. Noncompliant Items
The Subcontractor shall remove and replace, at its expense, all items that are not in compliance with the Specification requirements.
- D. Function-oriented commissioning includes the following services, which are to be provided by BMS specialist:
- verification of the external connections of the equipment
 - verification of the data transfer channels of the system
 - loading and testing of all basic and user programs belonging to the equipment
 - optimisation of the control parameters
- E. Point to Point Checkout – each I/O device (both field mounted as well as those located in FIP's) shall be inspected and verified for proper

installation and functionality. A checkout sheet itemizing each device shall be filled out, dated and approved by the Design Consultant for submission to the Client.

- F. Controller and Workstation Checkout – a field checkout of all controllers and front end equipment (computers, printers, modems, etc.) shall be conducted to verify proper operation of both hardware and software. A checkout sheet itemizing each device and a description of the associated tests shall be prepared and submitted to the Client at the completion of the project.
- G. System Acceptance Testing
1. Prior to acceptance testing, Subcontractor shall create, on an individual system basis, trend logs of input and output points, or have an automatic Point History feature for documentation purposes.
 2. An acceptance checklist shall be completed that documents compliance with each item of this Specification.
 3. All application software shall be verified and compared against the sequences of operation. Control loops shall be exercised by inducing a set point shift of at least 10% and observing whether the system successfully returns the process variable to set point. Record all test results and attach to the Operation and Maintenance Manual.
 4. Test each alarm in the system and validate that the system generates the appropriate alarm message, that the message appears at all prescribed destinations (workstations or printers), and that any other related actions occur as defined (i.e. graphic panels are invoked, reports are generated, etc.). Submit a test results Sheet to the Design Consultant.
 5. Perform an operational test of each unique graphic display and report to verify that the item exists, that the appearance and content are correct, and that any special features work as intended. Submit a test results sheet to the Design Consultant.
 6. Perform an operational test of each third party interface that has been included as part of the BMS. Verify that all points are properly polled, that alarms have been configured, and that any associated graphics and reports have been completed. If the interface involves a file transfer over Ethernet, test any logic that controls the transmission of the file, and verify the content of the specified information.

3.05 TRAINING

- A. All training shall be by the Subcontractor and shall utilize specified manuals and As-Built Documentation
- B. Operator training shall include not less than total seven sessions each of six-hour encompassing:
 - 1. Modifying text and graphics
 - 2. Sequence of operation review
 - 3. Selection of all displays and reports
 - 4. Use of all specified OS functions
 - 5. Use of portable operators terminals
 - 6. Trouble shooting of sensors (determining bad sensors)
 - 7. Password assignment and modification
- C. The training shall be undertaken in two phases. One training session shall be conducted at system completion, and the other session shall be conducted in accordance to approved scheduled training program.

END OF SECTION